DATA BASES – PROF. D. M. BRAGA – A. A. 2021/22

CALL 1 – JANUARY 28TH 2022 – DURATION: 1h 30m

SQL (24 points)

The schema describes time-based competitions of the Winter Olympic Games. Each Olympic Sport (*say, Alpine Skiing*) lists several Events (*say, men's SuperG, women's Slalom, …*), each assigning the corresponding medals. Events may have several competitions (*heats, semifinals, final*). Obviously, only one competition per event is the final (the one with flag ItIsTheFinal set to *true*). Disqualified is a flag to state whether a performance was not correctly completed (gates skipped, track boundaries trespassed, …); in this case the corresponding PerformedTime is to be considered invalid.

ATHLETE (AthleteID, Name, Country, Birthdate)

COMPETITION (CompID, Event, Date, ItIsTheFinal)

PARTICIPATION(AthleteID, CompID, PerformedTime, Disqualified)



1. Briefly but precisely explain what is extracted by the following queries [**5 p**]:

a.
```
select AthleteID
from PARTICIPATION
     natural join COMPETITION
where ItIsTheFinal = true
group by AthleteID
having count( distinct Event ) >= 3
```

b.
```
select Country, CompID
from ATHLETE A natural join PARTICIPATION natural join COMPETITION C
where ItIsTheFinal = true and Disqualified = false
group by Country, CompID
having count( * ) = 3
   and max( PerformedTime ) < ( select min( PerformedTime )
                                from PARTICIPATION natural join ATHLETE
                                where CompID = C.CompID and Disqualified = false
                                  and Country <> A.Country )
```

2. Extract in SQL:

- **a**. the Name of the youngest athlete of each Country; [**5 p**]
- **b**. the ID of the athletes who competed more than 4 times but never completed a try (always disqualified); [**5 p**]
- **c**. the number of gold medals won by each Country (athletes get a gold medal scoring the best time in a final). [**5 p**]

3. The following **materialized** view tracks the *personal best* of each athlete for each event:

```
create view PERBEST( Ev, AthID, BestTime ) as
    select Event, AthleteID, min( PerformedTime )
    from PARTICIPATION natural join COMPETITION
    where Disqualified = false
    group by Event, AthleteID
```

Assuming that performances are always inserted shortly after completion, and never deleted nor updated, formulate a set of triggers that incrementally maintain the materialization in response to *insertions* on table PARTICIPATION. Please note that only athletes with valid times for a specific event are listed in the view. [**4 p**]

Conceptual Design (8 points)

4. Design an E-R model for the database described in the following scenario:

A database stores the information about several boat-rental companies in a tourist harbour. For each company, we know the VAT number, the manager name, and the year it started its business. Each rental company manages a set of sail- and power-boats, each with a serial number, a description (type, size, and other optional specs), and the daily renting cost. For powerboats we also store the engine power and the tank size. Each single rental of each boat is tracked, with the corresponding start and end dates, boat identifier, and price paid. Customers can request to rent several boats in parallel, or one boat for several subsequent time periods. For every request that is satisfied, a contract is issued, and we store the fiscal code, name and surname of the customer, the code of the receipt, and the details of each single rental within the contract (which boat is rented, the number of days, and the cost).

- *The project is evaluated for completeness, correctness, legibility, minimality and self-explanatory nature.*
- *Remember to specify the **identifiers** of all entities and the **cardinalities** of all associations.*